

1. Administrative & Study Identification

Full Study Title: Study of Patritumab Deruxtecan Plus Pembrolizumab With Other Anticancer Agents in Participants With High-Risk Early-Stage Triple-Negative or Hormone Receptor-Low Positive/HER-2 Negative Breast Cancer (MK-1022-010, HERTHENA-Breast-03) **Short Title/Acronym:** HERTHENA-Breast-03 / MK-1022-010 **Protocol Number:** MK-1022-010 **NCT Number:** NCT06797635 **Sponsor/Company:** Merck **Investigational Product:** Patritumab deruxtecan (HER3-DXd), Pembrolizumab, Paclitaxel, Carboplatin, Doxorubicin, Epirubicin, Cyclophosphamide (Trade Names: Cytosan, Endoxana; ATC Code: L01AA01) **Phase of Development:** PHASE2 **Medical Monitor:** Clinical Director, Merck

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety of the study treatments in Part 1 (including Adverse Events, Dose-Limiting Toxicities, and discontinuations due to AEs). **Secondary Objectives:** To evaluate if adding patritumab deruxtecan improves the Pathologic Complete Response (pCR) rate—specifically pCR-No Ductal Carcinoma in Situ (DCIS) Rate Using the Definition of ypT0 ypN0—and to evaluate Event-Free Survival (EFS) and Overall Survival (OS) in Part 2.

2.2 Background & Rationale Among patients with high-risk, early-stage Triple-Negative Breast Cancer (TNBC), neoadjuvant chemotherapy plus immunotherapy is the standard of care. However, this treatment regimen often causes serious long-term medical issues including secondary cancers and cardiomyopathy. Furthermore, patients who fail to achieve pCR face approximately a 3-fold risk of breast cancer recurrence. This study evaluates HER3-DXd, an antibody-drug conjugate (ADC) designed to seek out HER3—a protein often found in breast tumors—to deliver targeted chemotherapy with fewer side effects and improved efficacy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Standard of care chemotherapy + pembrolizumab) **Blinding:** Open-label **Randomization:** 1:1:1 (In Part 2 Randomized Phase)

3.2 Planned Enrollment & Duration Targeted Enrollment: 372 participants **Number of Sites:** Multicenter / Global **Estimated**

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Locally advanced, non-metastatic (M0) breast cancer, defined as combined primary tumor (T) and regional lymph node (N) staging per current AJCC criteria: cT1c, N1-N2; cT2, N0-N2; cT3, N0-N2; or cT4a-d, N0-N2. 2. Centrally confirmed diagnosis of breast cancer that is triple-negative or HR-low+/HER2- breast cancer treated according to the TNBC paradigm. 3. Hepatitis B surface antigen (HBsAg) positive participants are eligible if they have received HBV antiviral therapy for ≥ 4 weeks with an undetectable HBV viral load. History of HCV infection is eligible if viral load is undetectable. 4. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 to 1 within 28 days prior to allocation/randomization. 5. Left ventricular ejection fraction (LVEF) of $\geq 50\%$ or $\geq$ lower limit of normal (LLN) as assessed by ECHO or MUGA scan.

4.2 Exclusion Criteria 1. Uncontrolled or significant cardiovascular disease before randomization. 2. Clinically significant corneal disease. 3. Human immunodeficiency virus (HIV) infection with a history of Kaposi sarcoma and/or multicentric Castleman disease. 4. Prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent, or an agent directed to another stimulatory/co-inhibitory T-cell receptor. 5. Prior treatment, including radiation, systemic therapy, and/or definitive surgery for currently diagnosed breast cancer. 6. Prior treatment with an anti-HER3 antibody and/or ADC consisting of an exatecan derivative topoisomerase I inhibitor (e.g., trastuzumab deruxtecan). 7. Metastatic (Stage IV) breast cancer or cN3 nodal involvement. 8. Known additional malignancy progressing or requiring active treatment within the past 5 years. 9. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 10. History of (noninfectious) pneumonitis/interstitial lung disease (ILD) requiring steroids, current pneumonitis/ILD, or suspected ILD/pneumonitis that cannot be ruled out. 11. Active infection requiring systemic therapy. 12. Concurrent active HBV and HCV infection. 13. Clinically severe respiratory compromise resulting from intercurrent pulmonary illness.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Part 1 (Safety Lead-In):** HER3-DXd (dose to be determined) + pembrolizumab 200 mg Q3W for 12 weeks, followed by pembrolizumab 200 mg Q3W + paclitaxel 80 mg/m² QW + carboplatin AUC1.5 QW for 12 weeks. * **Part 2 (Randomized Phase - 3 Arms):** Arm A receives HER3-DXd + pembrolizumab followed by carboplatin/paclitaxel + pembrolizumab.

Arm B receives carboplatin/paclitaxel + pembrolizumab followed by HER3-DXd + pembrolizumab. Arm C (Control) receives carboplatin/paclitaxel + pembrolizumab followed by doxorubicin/epirubicin + cyclophosphamide + pembrolizumab. * **Adjuvant Phase:** Post-surgery, participants receive pembrolizumab 400 mg Q6W for ~30 weeks (5 cycles).

Route of Administration: Intravenous (IV) Infusion **Duration of Treatment:** 24 weeks of neoadjuvant therapy prior to surgery, followed by ~30 weeks of adjuvant therapy.

5.2 Concomitant & Prohibited Therapy Permitted: Treatment of Physician's Choice (TPC) such as capecitabine or olaparib (for BRCA mutated) is permitted in the adjuvant setting for participants with residual disease. Postoperative radiotherapy is permitted if clinically indicated. **Prohibited:** Concurrent experimental anti-cancer agents.

6. Study Timeline & Visit Schedule

| Visit ID | Window | Key Activities/Procedures | | :--- | :---
| :--- | | **Screening** | Days prior to Day 0 | Informed Consent, Medical History, Central pathology confirmation of TNBC/HR-low status, baseline imaging. | | **Neoadjuvant Phase** | Weeks 1 to 24 | IV Infusions: Q3W (Pembrolizumab/HER3-DXd) and QW (Paclitaxel/Carboplatin); Safety labs, efficacy assessments. | | **Surgery** | 3 to 6 weeks post-neoadjuvant | Breast cancer resection; primary pathological complete response (pCR) assessment. | | **Adjuvant Phase** | Post-Surgery (~30 weeks) | Adjuvant pembrolizumab Q6W infusions, safety monitoring. | | **End of Study** | Follow-up | Long-term disease recurrence/survival follow-up (Estimated Study Completion 2034). |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Part 1:** Number of Participants Experiencing an Adverse Event (AE). 2. **Part 1:** Number of Participants Who Experience One or More Dose-Limiting Toxicities (DLTs). 3. **Part 1:** Number of Participants who Discontinued Study Treatment Due to an AE.

7.2 Secondary Outcomes 1. **Part 2:** pCR-No Ductal Carcinoma in Situ (DCIS) Rate Using the Definition of ypT0 ypN0. 2. **Part 2:** Event-Free Survival (EFS). 3. **Part 2:** Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Evaluated in all-participants-as-treated (those receiving \$ge 1\$ dose of study intervention), specifically monitoring for reductions in standard chemotherapy

toxicities. An AE is defined as any unfavorable and unintended sign, symptom, disease, or worsening of preexisting condition temporally associated with study treatment and irrespective of causality. **Dose-Limiting Toxicities (DLTs):** Defined by NCI CTCAE v5.0, assessed by investigator as drug-related (e.g., Grade 3 or 4 nonhematologic toxicity, Grade 3 or 4 febrile neutropenia, ILD, or prolonged delay >2 weeks in initiating Cycle 2). **Data Monitoring Committee (DMC):** Safety lead-in will be guided by a Bayesian Optimal Interval (BOIN) design to determine the optimal dose safely.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All-participants-as-treated set for safety; Intent-to-Treat (ITT) population for efficacy. **Sample Size Justification:** The study targets an enrollment of 372 participants, configured into a safety lead-in phase and a 1:1:1 randomized phase to detect statistically significant improvements in pCR and EFS/OS between the HER3-DXd arms and the active control arm. **Handling of Missing Data:** Standard survival analysis censoring for long-term endpoints and specific imputation methods for missing pathological data as per protocol.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical site monitoring, central pathology review of tumor tissues. **Audit Trail:** Standard Electronic Data Capture (EDC) system with audit trails and automated data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-1022-010 **Posting ID:** 587915306731761941 **Registry Number:** NCT06797635 **Sponsor/Lead Organization:** Merck **Study Title:** HERTHENA-Breast-03 **Official Regulatory Title:** An Open-label Randomized Phase 2 Study to Evaluate Safety and Efficacy of Patritumab Deruxtecan Plus Pembrolizumab Administered Either Before or After Carboplatin/Paclitaxel Plus Pembrolizumab Compared With Pembrolizumab in Combination With Chemotherapy Followed by Surgery and Adjuvant Pembrolizumab for High-Risk Early-Stage Triple-Negative or Hormone Receptor-Low Positive/ Human Epidermal Growth Factor Receptor-2 Negative Breast Cancer (HERTHENA-Breast03)

12. Clinical Framework (Breast Cancer Specific)

Disease Areas/Conditions: Breast Neoplasms, Breast Cancer
Preferred Name: Malignant neoplasm of breast **Preferred UMLS Name:** HER2-Negative Breast Cancer **Medical Codes:** SNOMED ID: 254837009 | MeSH Code: D001943 **Diagnosis Threshold:** Centrally confirmed TNBC or HR-low+/HER2- breast cancer; cT1c, N1-N2; cT2, N0-N2; cT3, N0-N2; or cT4a-d, N0-N2 **Medical Methodology:** Neoadjuvant antibody-drug conjugate (ADC) combined with immunotherapy, followed by surgery and adjuvant immunotherapy.

13. Study Procedures & Methodology

Management Pathway: Neoadjuvant systemic therapy → Surgery → Adjuvant systemic therapy **Education Protocol (HCP):** Protocol training on Bayesian dose-escalation design and HER3-DXd toxicity management **Education Protocol (Patient):** Informed consent, infusion schedules, and surgical planning **Quality Audit Mechanism:** Centralized pathology review for pCR assessment and independent safety reviews

14. Observation & Follow-Up Schedule

Total Duration: ~54 weeks of active treatment plus long-term survival follow-up **Onsite Visit Cadence:** QW and Q3W during Neoadjuvant; Q6W during Adjuvant **Remote Monitoring Frequency:** Routine telemedicine checks between active infusion cycles **Checklist Review Interval:** Prior to each Q3W or Q6W treatment cycle for AE assessments

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				